

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/train.csv')
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
```

```
5   Age          714 non-null   float64
6   SibSp        891 non-null   int64
7   Parch        891 non-null   int64
8   Ticket       891 non-null   object
9   Fare         891 non-null   float64
10  Cabin        204 non-null   object
11  Embarked     889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	89
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	3
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	4
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	1
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	3
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	51

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

df['Sex'].value_counts()

	count
Sex	
male	577
female	314

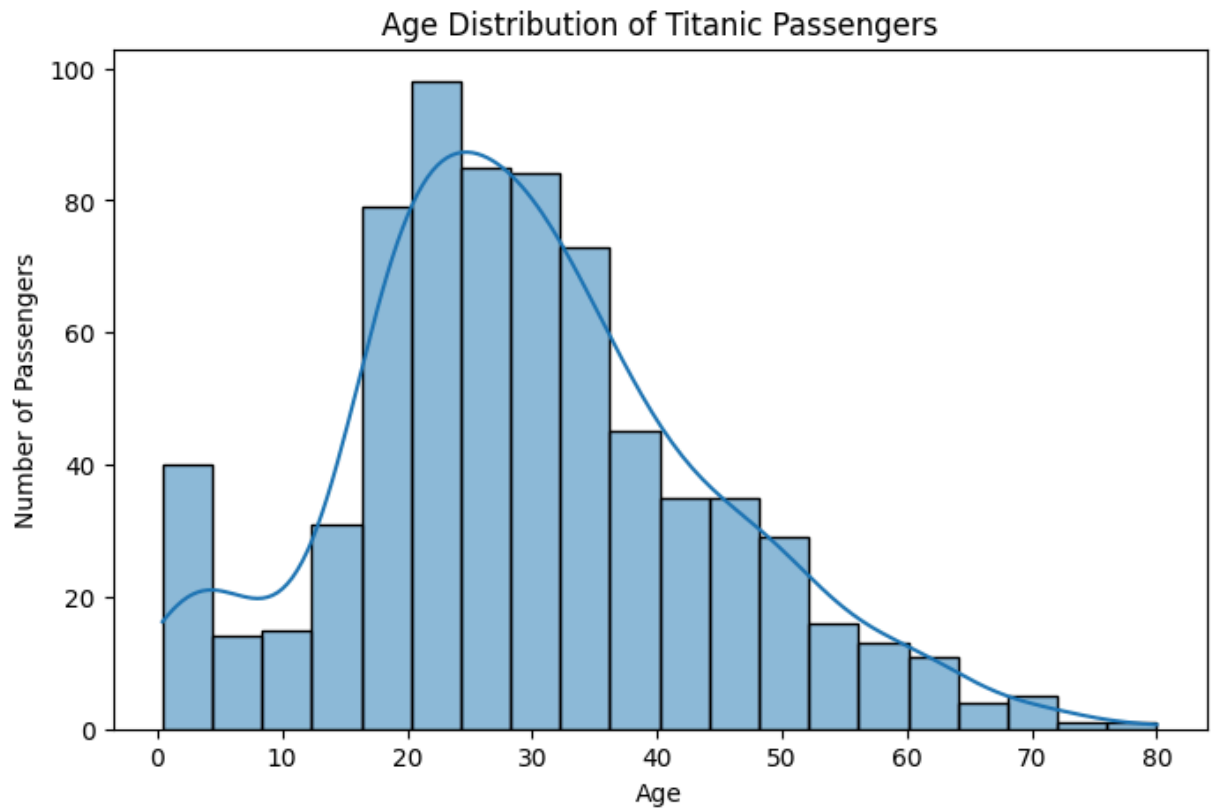
dtype: int64

df['Pclass'].value_counts()

	count
Pclass	
3	491
1	216
2	184

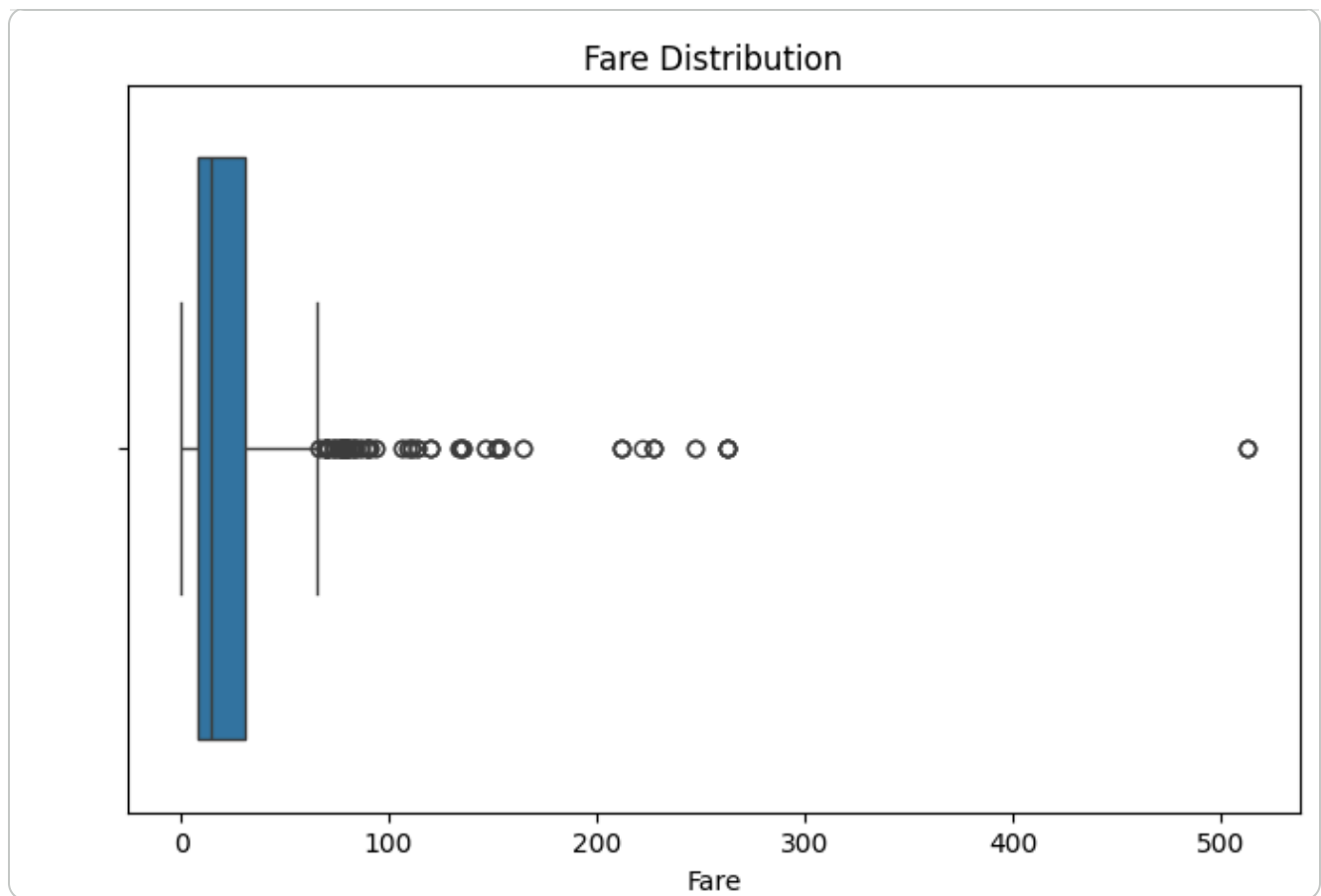
dtype: int64

```
plt.figure(figsize=(8,5))
sns.histplot(df['Age'].dropna(), kde=True)
plt.title('Age Distribution of Titanic Passengers')
plt.xlabel('Age')
plt.ylabel('Number of Passengers')
plt.show()
```



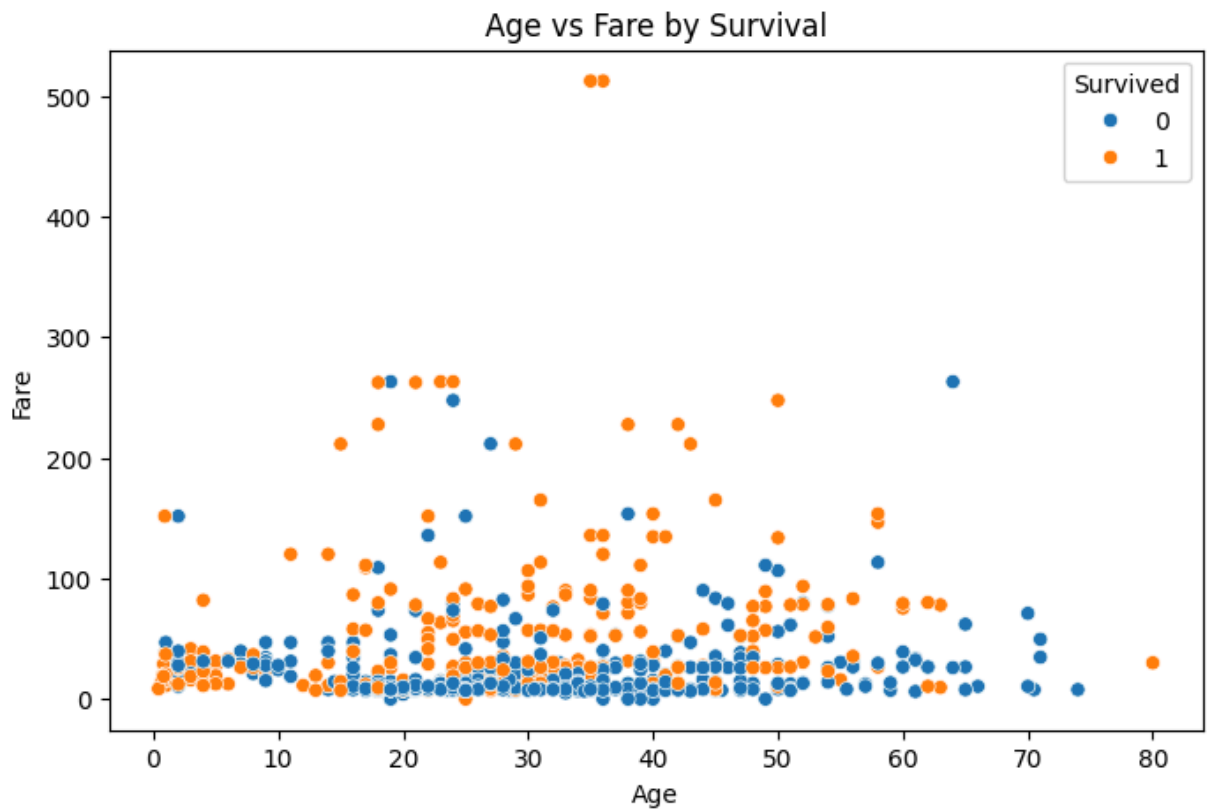
Observation: The histogram shows the distribution of passenger ages. Most passengers were concentrated in the younger and middle-age groups, while relatively fewer passengers were very old.

```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Fare'])
plt.title('Fare Distribution')
plt.xlabel('Fare')
plt.show()
```



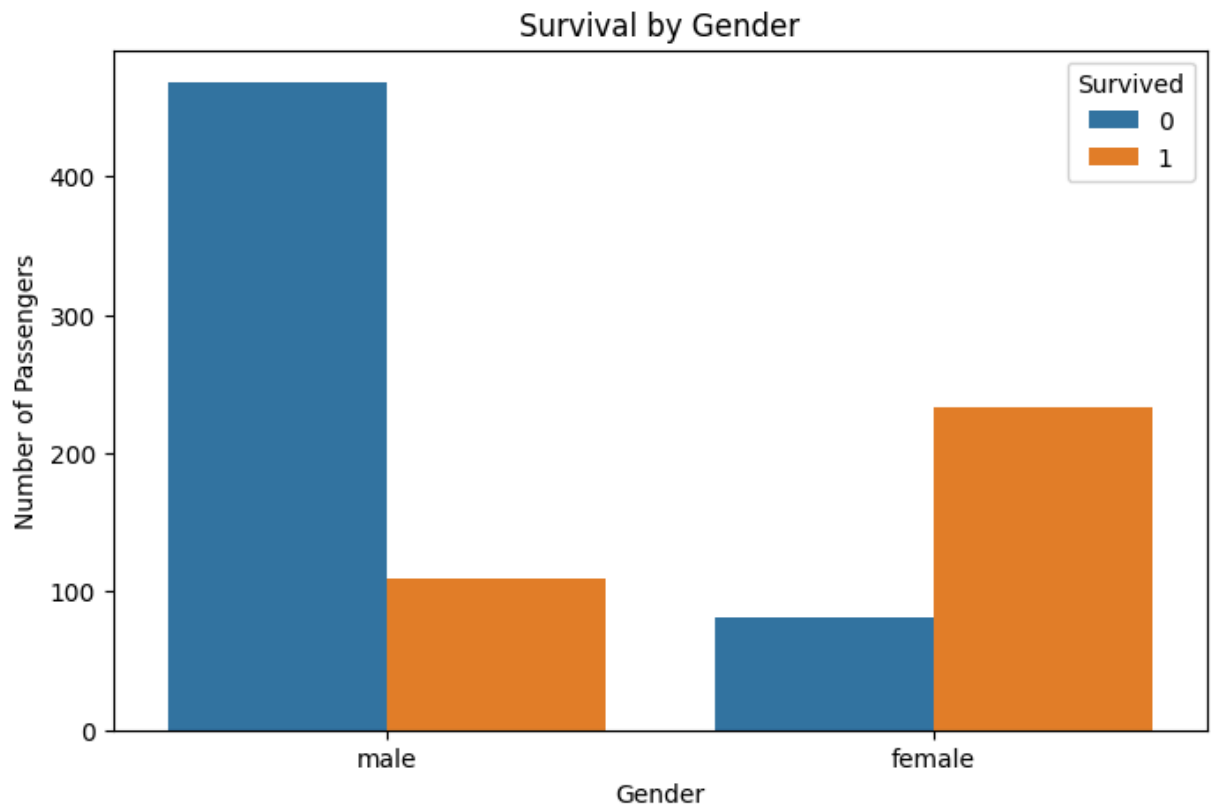
Observation: The boxplot shows that most passengers paid relatively lower fares, while some passengers paid considerably higher fares. These high fare values appear as potential outliers.

```
plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x='Age', y='Fare', hue='Survived')
plt.title('Age vs Fare by Survival')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```



Observation: The scatterplot shows the relationship between passenger age and fare. There is no strong direct relationship between age and fare, although some passengers paid very high fares.

```
plt.figure(figsize=(8,5))
sns.countplot(data=df, x='Sex', hue='Survived')
plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')
plt.show()
```



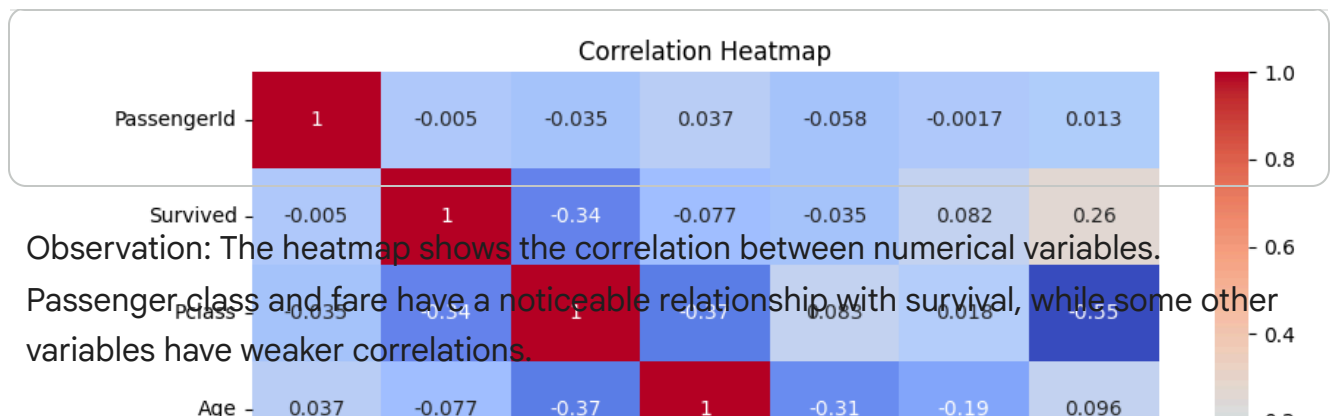
Observation: The chart shows a clear difference in survival between genders. Female passengers had a higher survival rate than male passengers in the Titanic dataset.

```
plt.figure(figsize=(10,6))

numeric_df = df.select_dtypes(include='number')

sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')

plt.title('Correlation Heatmap')
plt.show()
```



```
sns.pairplot(
    df[['Age', 'Fare', 'SibSp', 'Parch', 'Survived']].dropna(),
    hue='Survived'
)
plt.show()
```

